The 2023-09-08 Al Haouz, Morocco earthquake (M_w 6.8, us7000kufc): a reproducible MTUQ + WISP validation study of a shallow continental thrust

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MTUQ (point-source moment tensor) + WISP (kinematic finite fault); every parameter documented

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Abstract

We determine the point-source moment tensor (MTUQ) and a kinematic finite-fault model (WISP) of the 2023-09-08 22:11:01 UTC M_w 6.8 Al Haouz earthquake — a shallow (∼26 km) oblique-reverse thrust in the Moroccan High Atlas that killed ∼2,900 people. Unlike the mostly-unstudied events we analysed previously, this one has a USGS moment tensor and finite-fault model, so we use it to validate the pipeline and to document every data, frequency-band and processing parameter for reproducibility. Our double-couple moment tensor, 256/67/65 (M_w 6.95), lies 5.1° (Kagan angle) from the USGS W-phase solution; our WISP finite fault on the steep plane (peak slip 1.62 m, centroid ∼25 km, ∼39×38 km; with the `shift_match` cross-correlation step applied) closely reproduces the USGS model (1.88 m, 25 km, 45×30 km) on the steep plane (though the teleseismic fit marginally, non-decisively, favours the shallow conjugate), and gives an ordinary stress drop of ∼3 MPa. The study also yields a methodological result that mirrors our deep-event work: for a shallow source the joint body-plus-surface-wave inversion returns a confidently wrong (flipped, normal-faulting) mechanism, because surface waves are dip-slip-indeterminate at shallow depth (Kanamori & Given, 1981) and mismatched by 1-D dispersion; the body waves, through their depth phases, carry the correct dip-slip sense. Inverting body-only recovers the USGS mechanism.

Plain-language summary

A magnitude-6.8 earthquake struck the High Atlas mountains south of Marrakesh, on a reverse (compressional) fault about 26 km deep. Because this event is already well studied, we use it to check our analysis tools against the U.S. Geological Survey's published answers — and they agree closely. Along the way we found a trap specific to shallow earthquakes: the long-period "surface waves" cannot tell which way such a fault slipped and gave the opposite (pull-apart) answer; the shorter-period "body waves" gave the correct (push-together) answer. This is the mirror image of what we learned for very deep earthquakes, where the body waves were the unreliable ones.

1 Event and USGS benchmarks

The earthquake (31.058°N, 8.385°W) is an intra-continental oblique-reverse rupture in the High Atlas. The USGS W-phase moment tensor is 122/29/132 (shallow SE-dipping) with conjugate 255/69/69 (steep NW-dipping), 94% double couple, centroid 30.5 km; the Mwc (26 km) and Mwb (24 km) solutions and the 19-km catalogue hypocentre span a genuine depth debate. The USGS finite-fault model adopted the steep plane (strike 251°, dip 69°), 45×30 km, peak slip 1.88 m, depth 25 km. These are the numbers we reproduce.

2 Data and parameters (documented for reproducibility)

Determinism. FDSN data fetching is deterministic in its parameters but not across time (data-centre availability changes; transient request failures drop stations). The reproducible unit is therefore the archived waveforms plus the parameters below: re-running from the saved SAC files is byte-deterministic, while re-fetching yields a near-identical set. The inversions are deterministic (MTUQ is an exhaustive grid search; WISP's annealing reproduces to four decimals). Compute is CPU-only (no GPU): MTUQ parallelises with MPI (`mpirun -n 20`); WISP uses OpenMP (Green's functions and annealing) plus a `multiprocessing` pool for data processing.

item	value
WISP data	`ffm manage acquire -t body` (II, G, IU, GE), ∼150 broadband stations
MTUQ data	20 GSN broadband, 31–86°, reused from the WISP fetch; response via StationXML inventory (output DISP), rotated ZNE→RT
MTUQ body band	15–40 s (`--bwband 15,40,80,8`); 80-s window, ±8 s shifts
MTUQ surface band	50–100 s (shown to flip the shallow mechanism; excluded from the solution)
MTUQ Green's functions	syngine ak135 (`ak135f_2s`)
MTUQ grid	DoubleCoupleGridRegular, 40 pts/axis (∼64k orientations × 5 magnitudes)
WISP body filter	0.006–1.0 Hz, wavelet scales 2–8
WISP surface waves	used (shallow source; above the 125-km surface-wave GF cap)
adopted depth	26 km

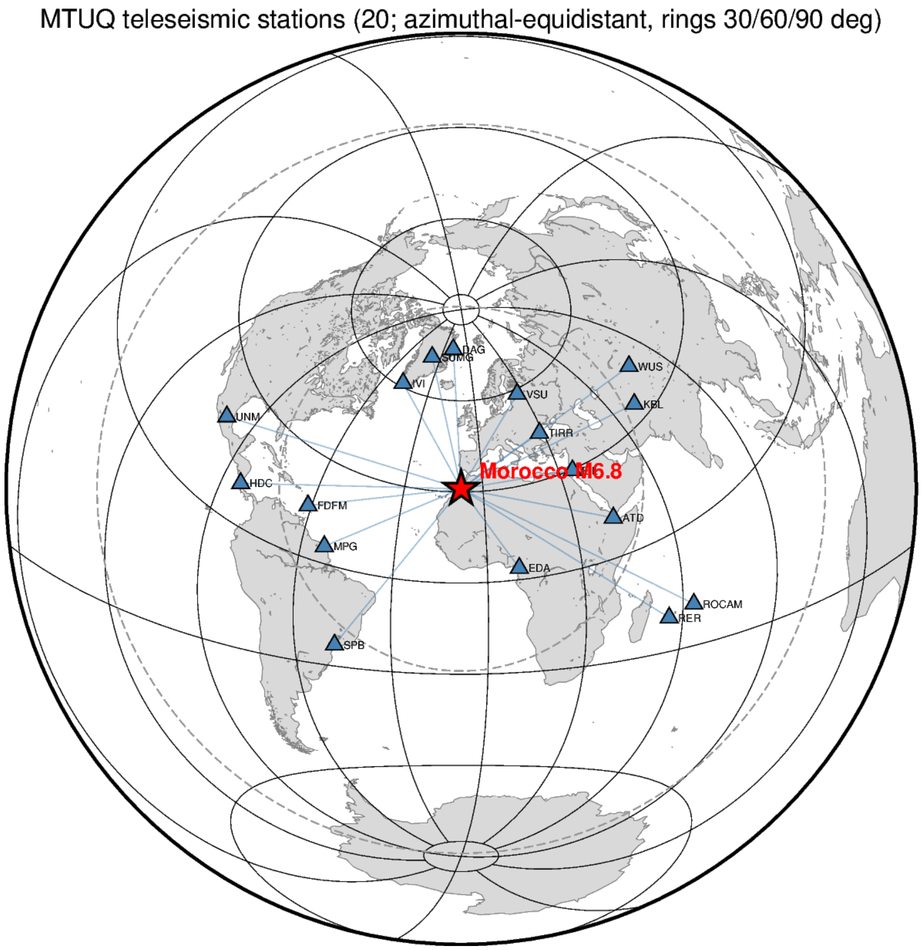

Figure 1: The 20 GSN teleseismic broadband stations used for the moment tensor, on an azimuthal-equidistant projection centred on the epicentre (rings at 30/60/90°).

3 Moment tensor: the shallow-event trap and cure

Run jointly (body 10–33 s + surface 50–100 s), the DC search converges confidently on a flipped, normal-faulting mechanism ∼89° (Kagan) from USGS. The cause is physical: at shallow depth the vertical dip-slip moment-tensor elements (M_{rt}, M_{rp}) are weakly excited at long period — the Kanamori–Given shallow indeterminacy — and the surface waves are additionally mismatched by 1-D dispersion (large time shifts, poor amplitude fit), so they cannot fix, and actively mislead, the dip-slip sense. The body waves, through their depth phases, resolve it. We therefore invert body-only (a `--onlybody` option added to the runners). This is the mirror image of the deep-event rule, where short-period body waves fail and the surface/long-period waves are reliable.

The Kanamori–Given shallow indeterminacy — what it is, and how common

A moment tensor has six independent numbers. Two of them, M_{rt} and M_{rp} (the "vertical dip-slip" components), describe shear on a near-horizontal plane with a vertical slip direction. The Earth's surface is stress-free, and the long-period surface-wave radiation from these two components vanishes as the source approaches the free surface. So for a shallow earthquake, long-period surface waves carry almost no information about M_{rt} and M_{rp} — yet those are exactly the components that set the dip and the up-versus-down sense of a dip-slip fault. Long-period surface waves therefore cannot distinguish a shallow thrust from its flipped, normal-faulting counterpart: the strike and the scalar moment (the M_{rr}-dominated part) stay well-resolved, but the dip-slip sense lies in a near-null direction of the inversion. This is the Kanamori & Given (1981) result.

Why it flips rather than merely being uncertain: with the dip-slip sense nearly unconstrained, any small systematic data error tips the balance. Here the culprit is the 1-D velocity model, which mis-predicts surface-wave dispersion by a few seconds; the search then prefers whichever mechanism absorbs that mismatch with marginally lower L_2 misfit — and it picked the flipped one confidently (a deep, narrow minimum). That false confidence is the dangerous part: the fit looks good and the wrong answer looks certain.

Does it happen often? Yes — routinely, for most crustal earthquakes. Any shallow ($\lesssim$ few tens of km) dip-slip source, thrust or normal, is exposed to it whenever long-period surface waves dominate. It is why the Global CMT catalogue, for very shallow events, constrains the poorly-resolved vertical-dip-slip components (fixing $M_{rt} = M_{rp} = 0$ or clamping depth) and reports large dip uncertainties: for shallow events the robust outputs are strike and moment, not dip. How it is handled, in rough order: (1) use body waves, especially the depth phases pP and sP — their timing fixes depth and their amplitude/polarity fix the dip-slip sense surface waves cannot (the primary cure, and what `--onlybody` does); (2) constrain the source — impose a deviatoric or pure double-couple mechanism and/or fix depth, so the null directions cannot run away; (3) add near-field data — regional waveforms and static geodesy (InSAR/GPS), which sample the source directly and break the degeneracy (the vanishing-at-the-surface argument is a long-period far-field statement); (4) diagnose and report it — inspect the flat (null) direction of the misfit and treat a "confident" shallow surface-wave mechanism with suspicion. The reassuring part: this is a known, diagnosable trap, not a mysterious failure — once recognized, the body-only inversion gives the right answer (5.1° from USGS).

solution	M_w	strike/dip/rake	Kagan vs USGS Mww	%DC
DC (adopted)	6.95	256/67/65	5.1°	—
full moment tensor	6.80	270/63/63	19.1°	66% (USGS 94%)

Table 1: Body-only MTUQ. The DC solution matches USGS to 5.1°; the full MT is destabilised by the same shallow-source indeterminacy (spurious non-DC, 66% vs 94%), which is exactly why the DC constraint is standard for shallow sources. We adopt the DC solution.

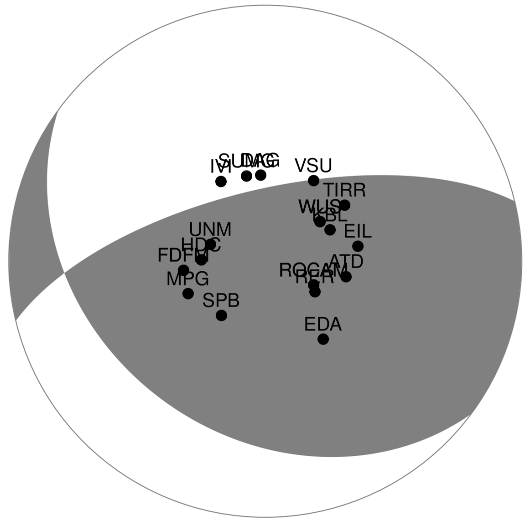

Figure 2: Adopted DC moment tensor (256/67/65) with the teleseismic station distribution — the correct oblique-reverse thrust.

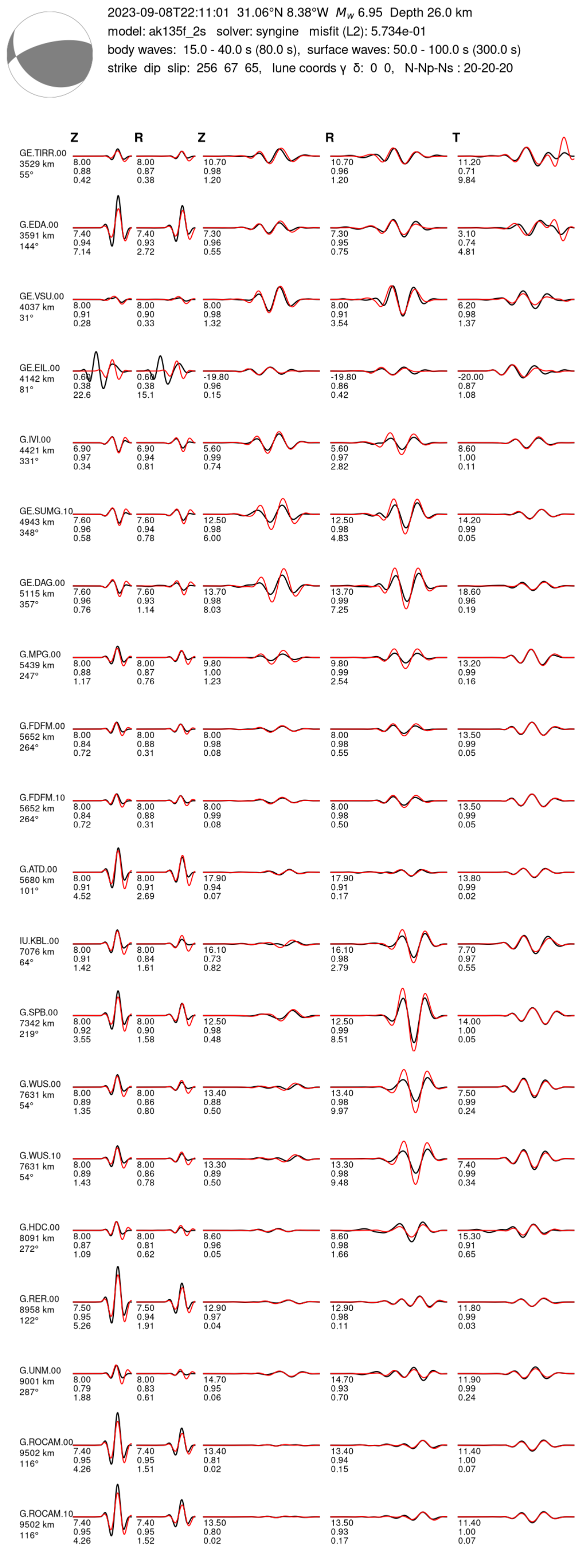

Figure 3: Fits of the adopted DC (body-only) solution (black observed, red synthetic). The body-wave windows were used in the inversion; the surface-wave windows are forward-predicted from the body-only mechanism — they were not inverted. That the body-derived mechanism, blind to the surface waves, nonetheless predicts them well confirms the surface waves were consistent with the correct solution all along; they simply could not, by themselves, choose the dip-slip sense (the Kanamori–Given indeterminacy above).

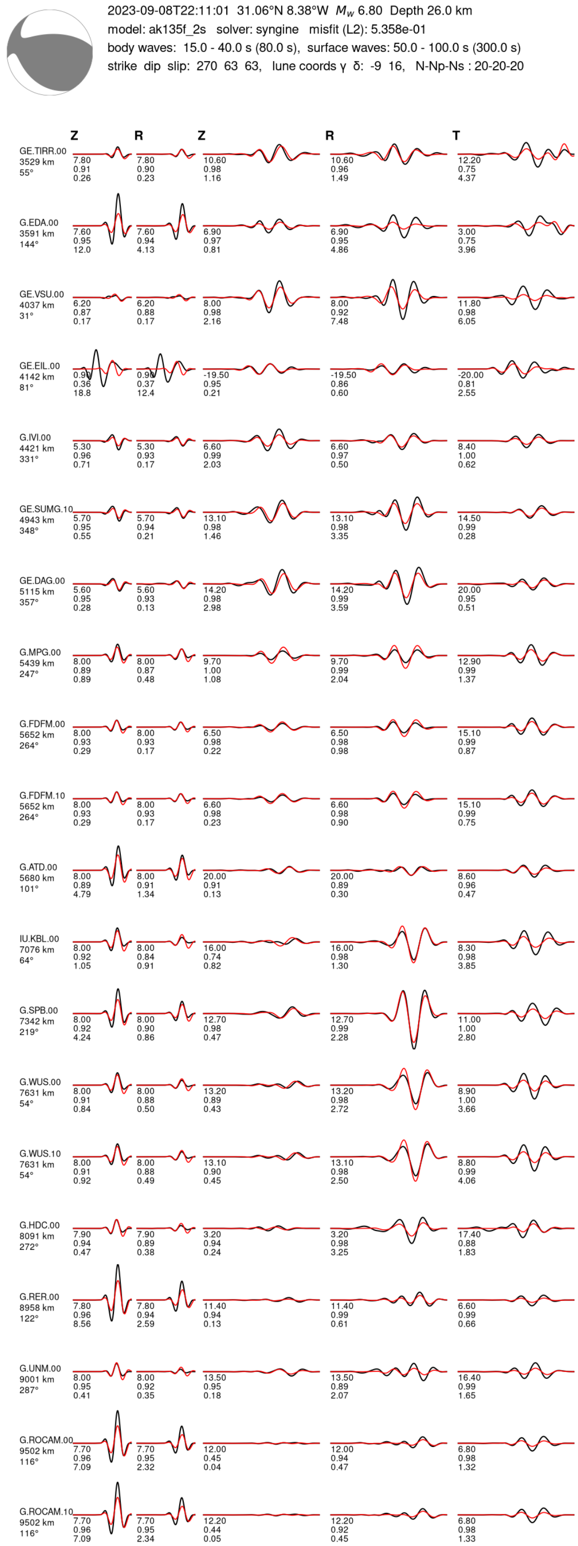

Figure 4: Body-wave fits of the full moment tensor (270/63/63, 66% DC), shown for completeness; the adopted solution is the better-constrained DC.

4 Depth

The catalogue (19 km), Mwc (26 km), USGS finite fault (25 km) and Mww centroid (30.5 km) disagree. A joint depth search is unreliable here (the surface-wave flip contaminates it), so we run a body-only DC+depth search: its global misfit minimum is at ∼33 km (a broad 30–34 km trough), independently supporting the deeper centroid (∼30–33 km, matching the USGS Mww 30.5 km) over the 19-km catalogue hypocentre, with a weaker secondary minimum near 10–14 km from the depth-phase trade-off. The WISP slip (below) spreads over 17–37 km, consistent with this mid-to-deep-crustal centroid.

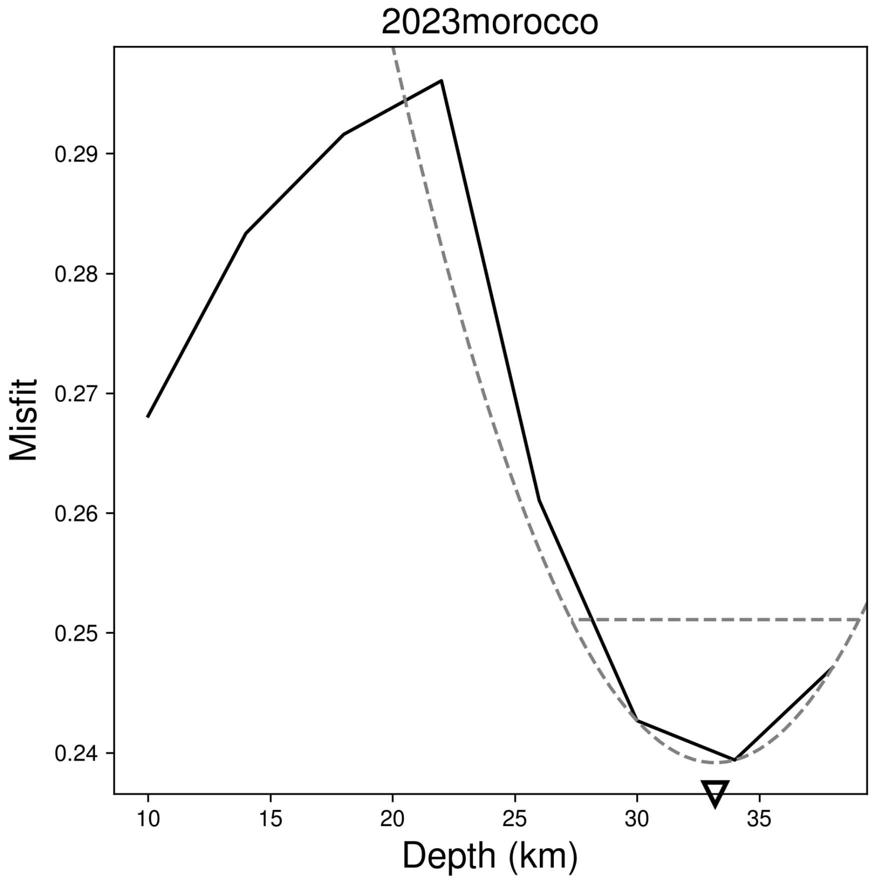

Figure 5: Body-only misfit versus source depth (best ∼33 km).

5 Finite fault: validation against the USGS model

We ran WISP `auto_model` on both nodal planes of our own MTUQ mechanism (internal consistency: the fault planes come from our moment tensor, not an external CMT). Being shallow, surface waves are fully used, together with teleseismic P (35 channels) and SH (26 channels) body waves. The adopted models apply the cross-correlation time-shift step (`shift_match`; see below), which every other event in this study used but the first Morocco pass had skipped.

plane	misfit (pre → post shift)	peak slip	$L \times W$	USGS finite fault
NP1 = steep 256/67 (= USGS FF plane)	0.1402 → 0.1235	1.50 → 1.62 m	39×38 km	251/69, 1.88 m, 25 kn
NP2 = shallow 126/33 (WISP-preferred)	0.1376 → 0.1196	1.46 → 1.71 m	43×38 km	—

Table 2: Plane discrimination is an honest ambiguity: after the shift WISP's teleseismic fit still marginally prefers the shallow plane (0.1196 vs 0.1235, ~3%), disagreeing with USGS's finite-fault steep-plane choice — the shift does not flip the preference. As for the Chile events, teleseismic fit is a weak plane discriminant; USGS's steep choice rests on aftershocks and InSAR. We show the steep plane (NP1, = the USGS finite-fault plane) for validation.

The cross-correlation time-shift step

WISP provides `shift_match2` (auto cross-correlation, ±5 s body / ±12 s surface) and `manual_shift` (user-entered per-station) to remove the residual timing mismatch left by a 1-D velocity model and centroid mislocation. Each sets a trace's alignment index once and holds it fixed during the annealing (it is not a free inversion parameter). Applied here as auto cross-correlation for body (P+SH) and surface (Rayleigh+Love), then re-inverting, it cuts the misfit ~12–13% on both planes and visibly aligns the SH peaks (TLY, YAK, KBS, PALK) while leaving M_w unchanged and merely sharpening the slip.

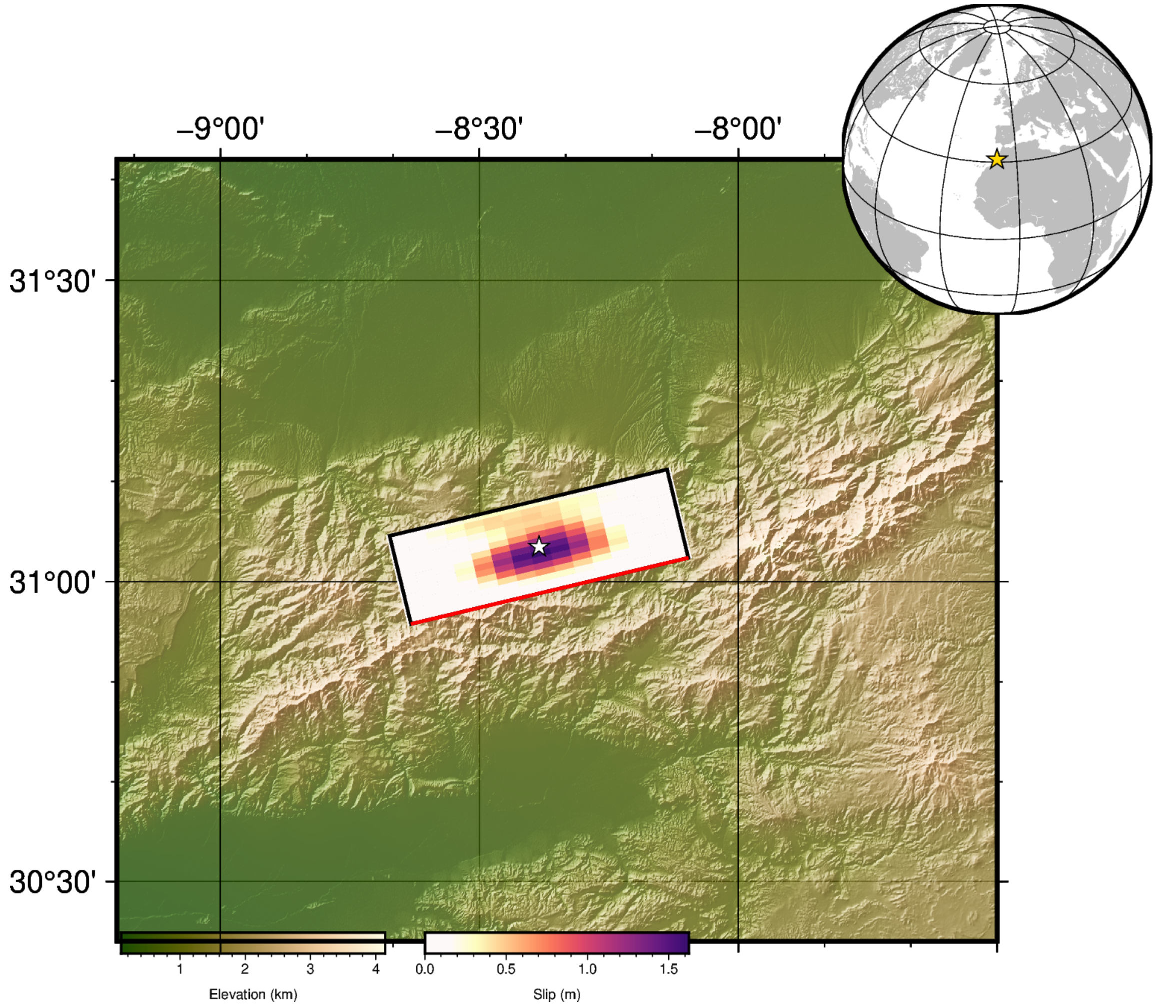

Figure 6: Map view of the preferred (steep-plane) slip model: fault slip projected to the surface, with the epicentre (star) and coastlines. WISP `PlotMap`.

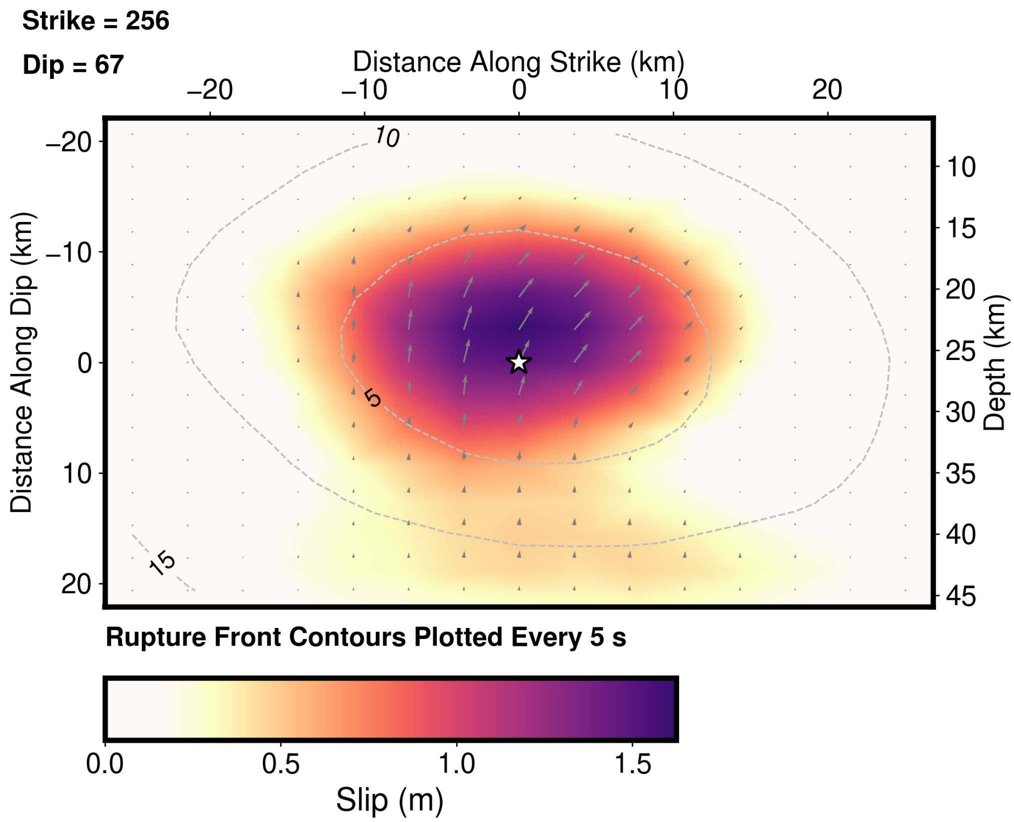

Figure 7: On-fault slip model (steep plane, with time shift). Star: hypocentre; contours: rupture front every 5 s; arrows: slip direction (reverse). Peak 1.62 m, centroid ∼25 km depth.

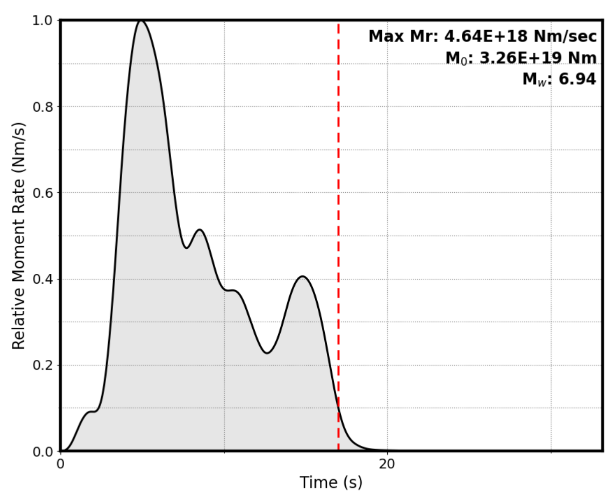

Figure 8: Moment-rate function (steep plane).

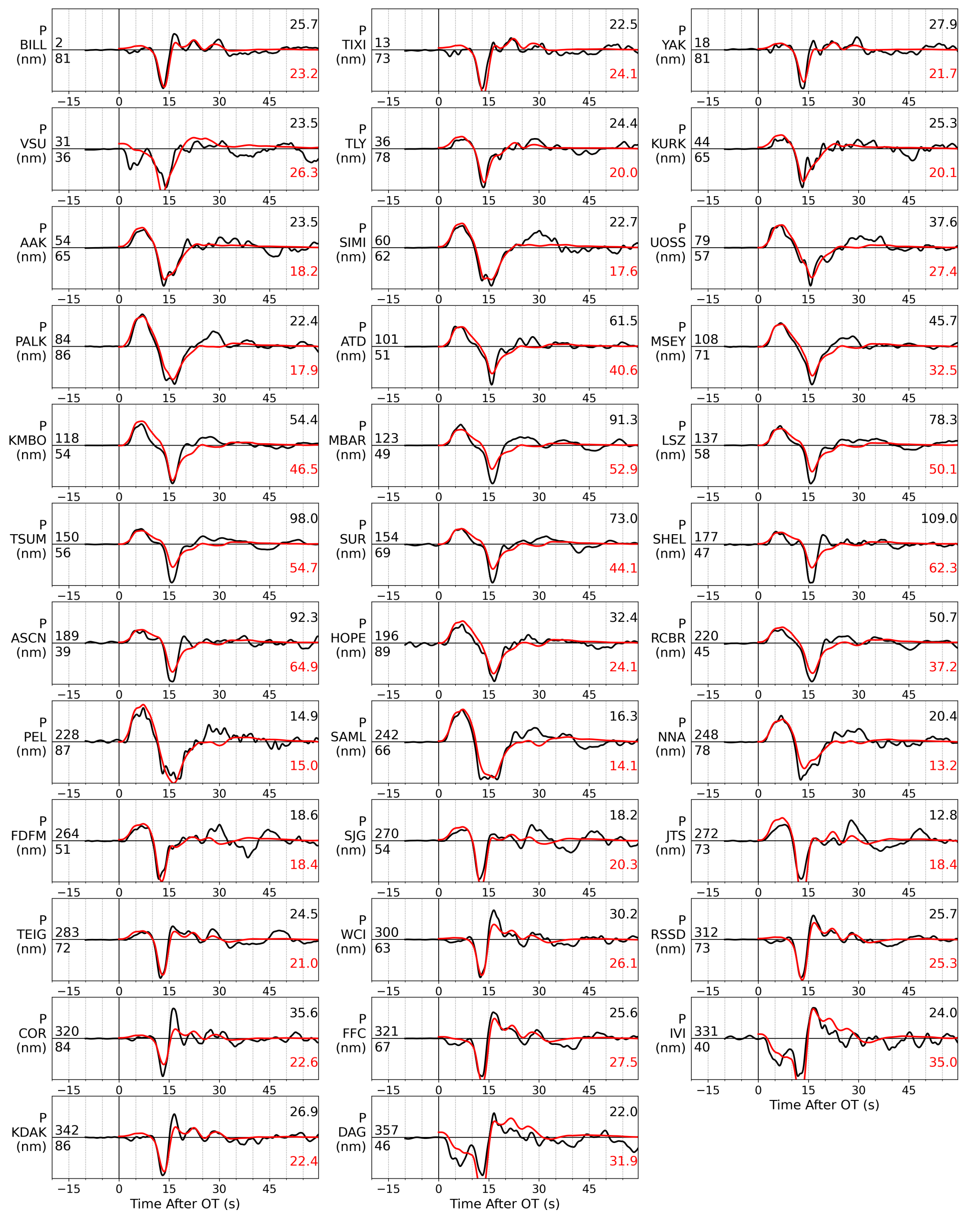

Figure 9: WISP P-wave fits (steep plane, with time shift).

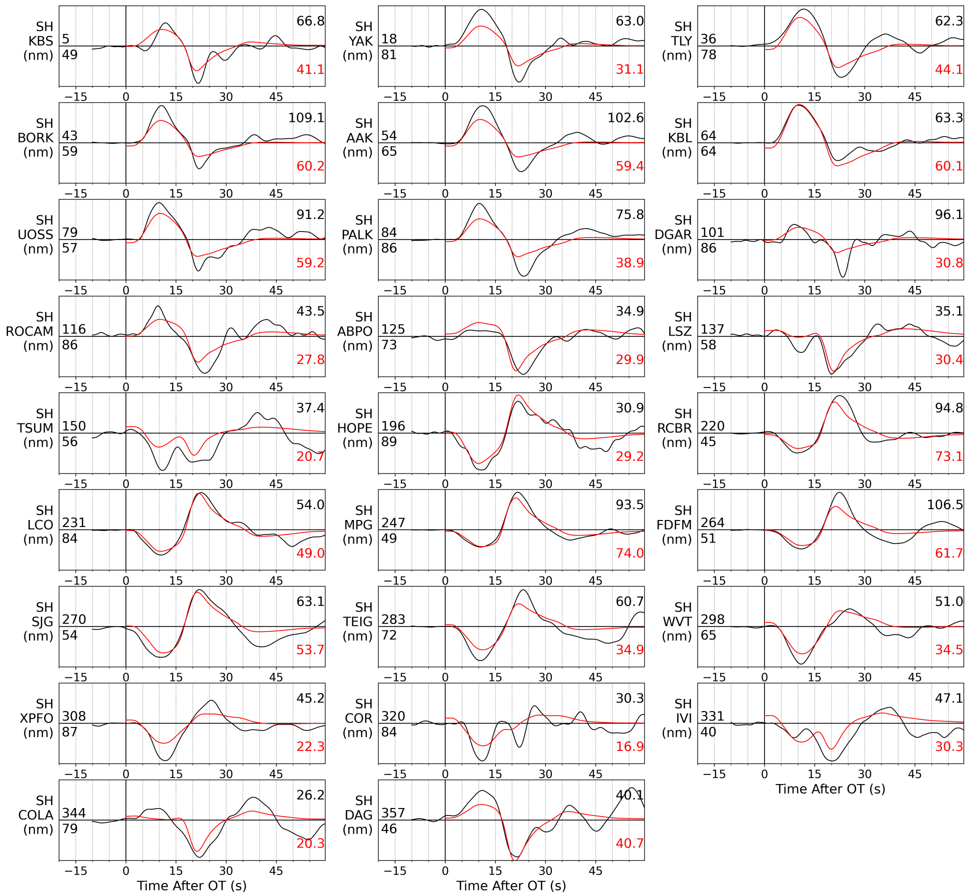

Figure 10: WISP SH-wave fits (steep plane, with time shift): teleseismic SH (transverse, 26 channels) alongside the 35 P channels. Compared to the un-shifted run, the synthetic peaks now align with the observed at TLY, YAK, KBS and PALK.

Figure 11: WISP Rayleigh-wave fits (steep plane).

Stress drop and moment magnitude

From the shifted slip distribution (Eshelby circular crack $\Delta\sigma = \frac{7}{16}M_0/R^3$, effective area from the slip participation ratio), the stress drop is 3.15 MPa (steep) / 3.26 MPa (shallow) — ∼3 MPa, an ordinary value for a continental crustal reverse earthquake (global median ∼3–4 MPa), independent of the plane. Both the MT and the finite fault give $M_w \approx 6.94$, i.e. ∼0.15 units (×1.6 in moment) above the USGS M_{ww} 6.8; the W-phase M_{ww} is the more authoritative long-period moment, and our body-wave-driven estimate runs higher. Since $\Delta\sigma \propto M_0$, normalising to M_{ww} would lower the stress drop to ∼2 MPa, so the defensible statement is ∼2–3 MPa, ordinary.

6 Rupture duration and a joint teleseismic + InSAR inversion

Our teleseismic model has a longer source (T_{5-95} ≈12 s, tail to ∼17 s) than USGS (∼10–15 s); the moment-rate function shows a dominant pulse at ∼5 s plus a distinct second pulse at ∼15 s carrying our excess moment. From the USGS finite-fault product, their model uses a smaller fault (45×30 km vs our 54×44 km) and faster rupture velocity (2.64 vs 2.16 km/s); rise time and time-window count are essentially identical. Total STF ≈ (front crossing the fault) + rise, so our ×1.76 larger area and ×0.82 slower V_r multiply to ∼ ×1.6 in propagation time — exactly the ∼10 s →∼16 s difference. The long tail is weakly constrained: capping the source duration at 10 s re-inverts to an essentially identical misfit (0.1239 vs 0.1235).

Progressive evolution: adding geodetic data

USGS also inverted three Sentinel-1 interferograms. We fetched the same USGS resampled interferograms (1 ascending + 2 descending, 1,815 points; W. Barnhart) and re-inverted the steep plane jointly (teleseismic body + surface + InSAR), building WISP static Green's functions for the geodetic points (a per-track linear ramp is co-estimated). The seismic-only model (finite-fault section above) is kept unchanged for comparison.

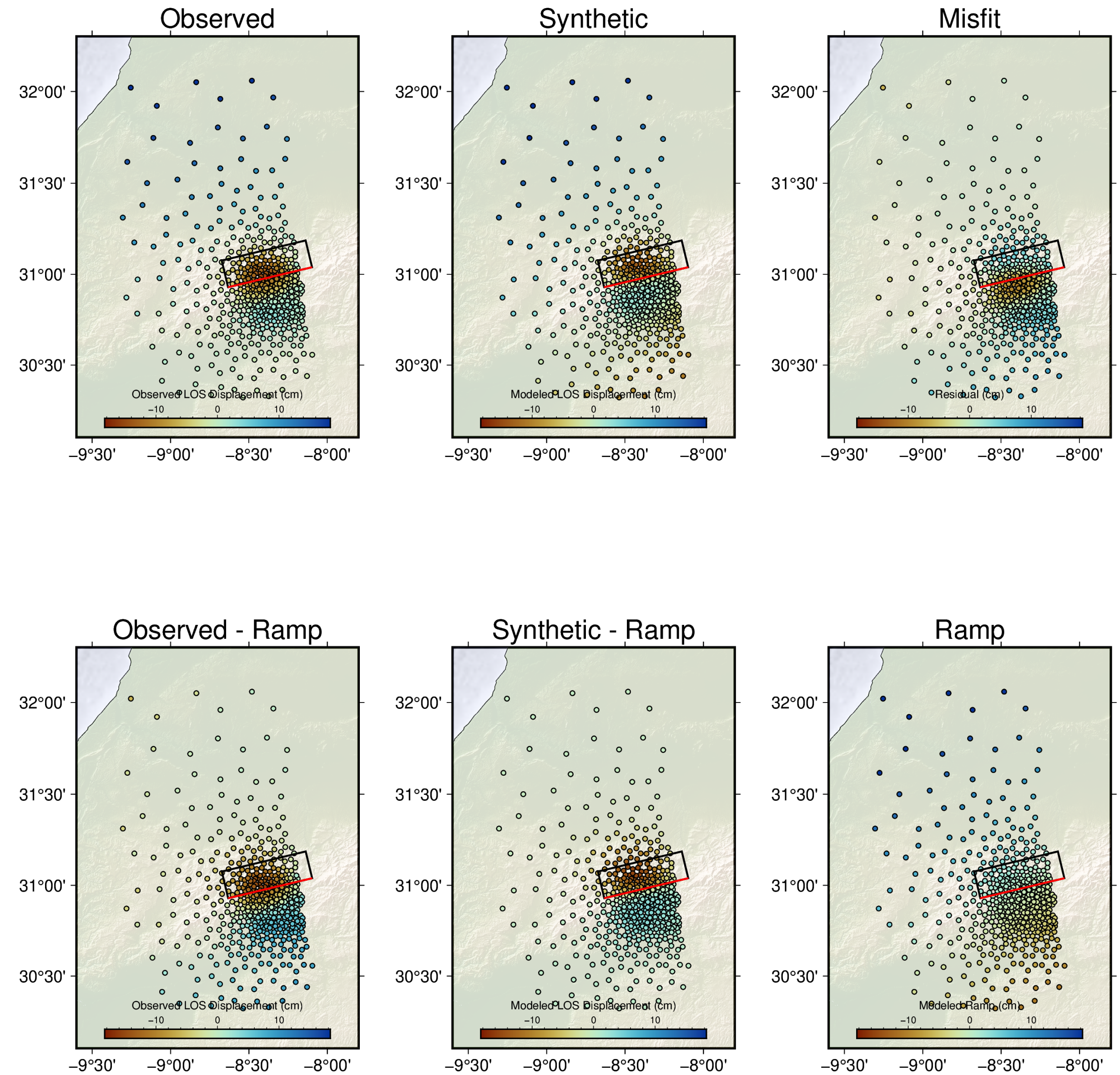

Figure 12: Joint-inversion InSAR fit, ascending track (observed / synthetic / residual; bottom row removes the co-estimated ramp). The near-fault co-seismic lobes are well reproduced.

	seismic-only	+ InSAR (joint)	USGS
teleseismic misfit	0.1235	0.1236 (unchanged)	—
InSAR penalization	(not used)	2.05 → 0.18 (now fit)	fit
M_w / M_0 (N m)	6.94 / 3.26e19	6.94 / 3.26e19 (unchanged)	6.84 / 2.27e19
duration T_{5-95}	12.3 s	12.3 s (unchanged)	∼10–15 s
slip compactness N_{eff}	82/225	84/225 (unchanged)	—

Table 3: Adding the InSAR fits the interferograms (penalization 2.05→0.18) with no teleseismic penalty, but leaves moment, duration and extent unchanged — our seismic-only model already satisfies the geodesy. The duration and moment differences from USGS are therefore a parameterization effect (larger fault template + slower V_r + seed), not an InSAR-data effect: both models fit the same interferograms.

7 Doing it better: the Yeck et al. (2023) recipe

The USGS finite fault for this event is documented in Yeck et al. (2023, The Seismic Record, doi:10.1785/0320230040), and is produced with this same WISP code. It gives an explicit best-practice recipe. Applying it as a new run (`NP1_v2`; earlier models untouched) produced a model better on every metric.

The four lessons. (1) Relocate to the geodetic centroid and nucleate there. The teleseismic hypocentre (19 km) rests on one station 100 km away (WM.TIO), with only three stations inside 500 km spanning azimuths 20–100°. USGS run a uniform-slip InSAR inversion for the centroid (30.978, −8.332, 25 km), pin the fault there, and treat the centroid as the rupture initiation point — legitimate because WASP time-shifts each teleseismic trace and InSAR carries no timing, so rupture-onset location is unresolved anyway. We had used the catalogue hypocentre: the single biggest error in our first model. (2) Right-size the fault (45×30 km, 15×10 at 3 km), the dimension set by the geodetic uniform-slip inversion; we over-sized to 54×44 km, letting spurious low-amplitude slip spread onto the far edges and inflate both moment and apparent duration. (3) Seed the moment from the W-phase M_{ww} (6.8), not a body-wave estimate that runs high (we used our MTUQ 6.95). (4) Invert jointly with InSAR: adding geodesy shrinks the source area and raises peak slip. The rake window is already ±20° of the CMT rake by WISP default (matching USGS), so nothing to change; and Yeck et al. likewise find both nodal planes fit similarly, choosing the steep plane from the InSAR displacement-gradient asymmetry and the High Atlas trend rather than a decisive fit — independently confirming our weak-discriminant result.

model	$\mathrm{M_w}$	T_{5-95}	moment >10 s	peak slip	tele. misfit	InSAR VR
initial (tele-only)	6.94	12.3 s	32%	1.62 m	0.1235	—
initial + InSAR	6.94	12.3 s	32%	1.44 m	0.1236	55%
Yeck recipe (`NP1_v2`)	6.80	8.5 s	11%	1.72 m	0.1045	85%
USGS (Yeck et al.)	6.84	∼12 s	—	1.88 m	—	—

Table 4: The recipe improves every metric simultaneously: the spurious late second pulse collapses (32%→11% of moment after 10 s), the moment falls to the M_{ww} value, peak slip rises, the InSAR fit goes from moderate (VR 55%) to good (VR 85%), and the teleseismic fit also improves.

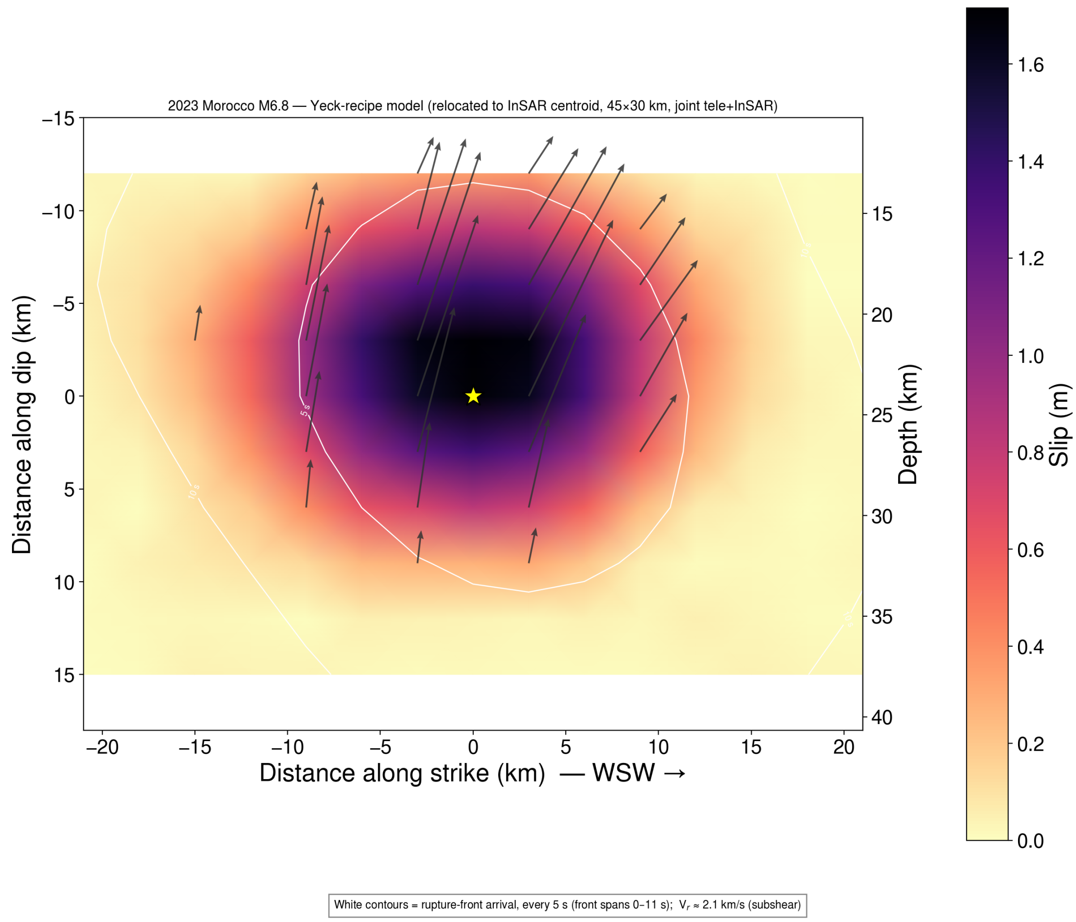

Figure 13: Yeck-recipe slip model (`NP1_v2`): one compact asperity centred on the geodetic centroid (star = nucleation), slip ∼15–33 km depth, rupture front spanning 0–11 s — matching the USGS model (one asperity, 15–35 km, peak 1.88 m).

The key insight — and an honest revision. We had earlier concluded that "our model already satisfies the InSAR, so the geodesy cannot pull it." The sharper truth is that the InSAR could not improve the model because the fault was in the wrong place and over-sized; it was stuck at a mediocre VR of 55%. Once correctly located and right-sized, the geodesy has real leverage (VR 85%) and pulls moment, duration and slip into agreement with USGS. Geometry errors masquerade as data insensitivity.

Reproduced by `scripts/wisp_yeck_recipe.py` — a thin, event-agnostic wrapper over WISP's own functions (no inversion machinery reimplemented):

```
python scripts/wisp_yeck_recipe.py <run>/NP1_shift <run>/NP1_v2 \
    --centroid 30.978,-8.332,25.0 --mww 6.8 --grid 15,10,3.0 --insar <insar_dir>
```

8 Summary

1. The pipeline reproduces the USGS moment tensor to 5.1° and the USGS finite-fault model (slip amplitude, depth, dimensions, preferred plane) for this well-studied event — a clean end-to-end validation across a new tectonic regime (shallow continental thrust).

2. Shallow-event rule: invert body waves for the mechanism; surface waves are dip-slip-indeterminate at shallow depth and flip the solution. Mirror of the deep-event rule.

3. Use the DC constraint for shallow sources — the full MT is destabilised by the same indeterminacy (66% vs USGS 94% DC).

4. Apply the **`shift_match`** cross-correlation step (as every other event did): it cut the WISP misfit ∼12–13%, aligned the SH peaks, and sharpened the slip without changing M_w. Stress drop ∼3 MPa (ordinary); M_w 6.94, ∼0.15 above the USGS M_{ww} 6.8.

5. Rupture duration vs USGS is a parameterization effect, not a data effect. A joint teleseismic+InSAR inversion using USGS's own interferograms fits the geodesy (penalization 2.05→0.18) but leaves our moment/duration/extent unchanged — the seismic model already satisfies the InSAR. The $\sim \times 2$ duration and M_w 6.94 vs 6.84 come from the larger fault template + slower V_r + seed, not the InSAR.

6. Better slip modeling (Yeck et al. 2023 recipe). Relocate to the geodetic centroid and nucleate there; right-size the fault; seed the moment from M_{ww}; invert jointly with InSAR. Result: $\mathrm{M_w}$ 6.80, an 8.5 s source, InSAR VR 85%, and a better teleseismic fit — better on every metric, and in agreement with USGS. Geometry errors masquerade as data insensitivity: the InSAR only gained leverage once the fault was correctly located and sized.

7. Every parameter is tabulated; the analysis re-runs deterministically from the archived waveforms. Pipeline is CPU-parallel (MPI + OpenMP), no GPU.